

## RESEARCH ARTICLE

# Multi-Stage Adversarial Defense for Online DDoS Attack Detection System in IoT

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**ABSTRACT** Machine learning-based Distributed Denial of Service (DDoS) attack detection systems have proven effective in detecting and preventing DDoS attacks in Internet of Things (IoT) systems. However, these DDoS attack-detection models are vulnerable to adversarial attacks. Adversarial attacks are small, crafted adversarial perturbations that are designed to fool machine learning model, leading misclassification of input. The existing defence technique primarily focuses on batch learning-based DDoS attack detection systems, that are not equipped to handle multiple and unknown adversarial attacks in real time. To address this challenge, a novel Multistage Adversarial Attack Defense (MSAAD) framework has been developed to protect online DDoS attack detection systems from adversarial attacks. The framework consists three defense layers: 1) the Resilient Adversarial Detector and Purification (RADP), which detects and purifies adversarial attacks targeting online DDoS attack detection systems against multiple and unknown adversarial attacks. 2) a multiple classifier, that increase complexity for an attacker to replicate the DDoS attack detection model. 3) the Multi-Armed Bandit (MAB) with Thompson Sampling, which dynamically selects the optimal classifier or ensemble of classifiers for each incoming traffic request. The experiment result shows the effectiveness of the proposed MSAAD framework using the IOTID20 and CICIOT2023 datasets. The accuracy of the MLBTSE based DDoS attack detection model improved from a range of 32.38%-60.58% to 99.39%-99.48% for the IOTID20 dataset and from range of 66.60%-86.20% to 99.01%-99.14% for the CICIOT2023 datasets, respectively, in the adversarial attack scenario.

**INDEX TERMS** IoT security, adversarial attacks, online learning, convergence analysis, adversarial detection, DDoS attack detection, online learning robustness.

## I. INTRODUCTION

The exponential growth of the Internet of Things (IoT) over the past decade has significantly transformed various sectors, including smart cities, energy grids, homes, and transportation. Ordinary devices have become sufficiently smart and intelligent for collaboration. This enables the interconnection of numerous IoT devices, thereby facilitating a seamless data exchange and automation. Despite the wide range of applications and technological developments, IoT remains vulnerable to cyberattacks [1]. A major weakness of IoT system is the lack of built-in security features. IoT

devices often lack strong built-in security measures and are typically deployed in different environments with limited computational resources, making them attractive targets for cyberattacks [2].

A DDoS attack is one of the most challenging forms of cyberattack due to the complexity and scale of IoT systems. DDoS attacks aim to overwhelm network resources, render IoT devices and services unavailable for legitimate users [3]. Such interruptions may have serious consequences, given the crucial role of IoT devices in various applications. For instance, a denial-of-service attack can interrupt the flow of critical patient data in the healthcare sector, leading to inaccurate or delayed medical responses [4]. Similarly, in industrial automation, DDoS attacks have the potential to

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stop automation of production lines, resulting in large financial losses. To combat these threats, DDoS attack detection systems have been used to protect against DDoS attacks and other types of cyber threats. DDoS attack detection systems frequently use machine learning models to detect DDoS attacks efficiently [4]. However, traditional DDoS attack detection system models are vulnerable to adversarial attacks that manipulate the input data to evade detection.

Adversarial attack was first identified in machine learning application within the field of computer vision [5]. The attack was performed by inserting small perturbations in the form of carefully crafted inputs to deceive machine learning model in order to misclassify the object. These attacks can significantly degrade the performance of DDoS detection, leading to incorrect detection and potential system failures [6]. Machine learning systems process input in two different phases: training and testing. During the training phase, data is fed into machine learning algorithms to develop a predictive model. In the testing phase, the trained model utilizes a separate testing dataset to make predictions or classifications. Adversarial attacks can occur during both phases. To enhance the robustness of machine learning models, researchers have developed various DDoS defense strategies applicable during both the training and testing phases [7].

Existing adversarial defense methods are designed for batch learning in DDoS attack detection. This approach is ineffective in resource-constrained, dynamic IoT environments, where concept drift detection and adaptation are critical. Furthermore, the current adversarial defenses are tailored to detect known adversarial attacks by training on predefined attack patterns

Defending IoT systems against adversarial attacks is complex in the context of online machine learning. Online machine learning can train models using continuously incoming IoT data streams in dynamic environments [8]. These models can reduce memory requirements for data storage and learn new data patterns by processing one data sample at a time. Additionally, online learning addresses issues related to concept drift and facilitates real-time processing [9]. The dynamic nature of online learning requires continuous defense mechanisms that can adapt to new types of adversarial strategies. Ensuring the robustness of online learning models against adversarial attacks is crucial for maintaining the security and reliability of IoT systems. Without adequate protection, these models can continuously learn from compromised data, leading to gradual degradation in performance and an increased risk of security breaches [10].

To address the challenges outlined above, this research paper focused on the following research questions: (1) What strategies can be employed to enhance DDoS attack detection in dynamic IoT environments? (2) What strategies can defend the proposed online DDoS attack detection system against multiple and unknown adversarial attacks? (2) What measurable improvements in DDoS detection performance and adversarial resilience can be achieved using IoT datasets?

These research questions guide the development of the proposed MSAAD framework, which addresses the current challenges in adversarial defense strategies for IoT systems.

MSAAD is proposed to defend an online DDoS detection model against adversarial attacks. It is a comprehensive framework designed to protect an online DDoS attack detection system from such threats. The MSAAD framework comprises three key stages: (1) Resilient Adversarial Detector and Purification (RADP), (2) multiple classifiers, and (3) an ensemble defense method. The RADP defense mechanism detects and purifies adversarial attacks targeting an online DDoS detection system, accommodating both multiple and unknown attacks. The framework consists of a Resilient Adversarial Detector (RAD) and Purification Modules. The RAD module detects adversarial attacks by leveraging the strengths of Generative Adversarial Networks (GANs) and multisource adversarial perturbations (MSAP). A GAN was used to reduce the gap between the training data and the real-world data. This makes the DDoS detector more aware of both the known and unknown adversarial attacks. MSAAP crafts various adversarial attack methods to create diverse perturbations to train an adversarial attack detector. This enhances its ability to detect a broader range of adversarial attacks and makes it more robust and resilient in real-world applications. The pruning method is used to purify adversarial perturbations to enhance the robustness of online learning models by systematically removing detected data points using RAD.

The multi-classifier defense method is designed to complicate the replication of the DDoS attack-detection model by adversarial attackers. These attackers cannot easily predict which model will classify a given request. By incorporating a diverse array of classifiers, the system introduces greater uncertainty, making it more challenging for adversarial strategies to succeed. Furthermore, the inclusion of more robust classifiers enhances overall performance, as the ensemble leverages the strengths of each model, resulting in a more resilient defense. This approach increases the system's resilience by introducing additional uncertainty when two or more classifiers are employed. The Multi-Armed Bandits (MAB) algorithm with Thompson Sampling (MABTSE) is utilized to dynamically select the optimal classifier or ensemble of classifiers for each incoming traffic request. MABTSE has been applied for online DDoS attack detection in conjunction with the AUWPAE ensemble approach [11].

The proposed MSAAD framework validated using two publicly available datasets: IoTID20 and CICIOT2023. The MSAAD framework effectively mitigates adversarial attacks while maintains high DDoS attack detection accuracy. The accuracy of the MLBTSE DDoS attack detection model improved from a range of 32.38%-60.58% to 99.39%-99.48% for the IOTID20 dataset and from a range of 66.60%-86.20% to 99.01%-99.14% for the CICIOT2023 datasets respectively, in the adversarial attack scenario.

The main contributions of this work are:

- 1) An online DDoS attack detection model utilizing a Multi-Armed Bandit (MAB) approach with Thompson Sampling (MLBTSE) that dynamically selects the optimal classifier or ensemble of classifiers for each incoming traffic request.
- 2) A novel Multistage Adversarial Attack Defense (MSAAD) framework designed to protect online DDoS attack detection systems from multiple and unknown adversarial attacks.
- 3) Validate the proposed framework using two publicly available IoT datasets and compared with other state-of-the-art online DDoS attack-detection models.

The rest of this paper is organized as follows. Section II presents related work on adversarial attacks and defense from cyberattack detection solutions. Section III describes the proposed MSAAD framework and MLBTSE online DDoS attack detection. Section IV presents experimental setup, dataset, and performance metrics. Section V presents discusses the results. Finally, Section VI presents the conclusion and paper future research direction.

## II. RELATED WORK

### A. ONLINE LEARNING

Online learning techniques are used to train models using continuously incoming IoT streams in dynamic IoT environments. IoT devices possess limited computing power and storage capacity [12].

This hinders their ability to handle and retain massive amounts of data and complex DDoS attack-detection models. Therefore, it is critical to develop detection models with low spatial and temporal complexities. Online learning models can reduce memory requirements for data storage and adapt new data patterns. Online learning models can often achieve real-time processing and address concept drift issues. Therefore, in dynamic environments, online learning is often more effective than batch learning. The batch learning method trains a learning model on the entire training dataset, in contrast to online learning methods. Online learning continuously updates the learning model with new data samples when it arrives within a short execution time. The online DDoS attack detection model often encounters concept drift issues when data distribution shift occurs. There are three major types of data distributions over time that cause concept drift: sudden, gradual, and recurring. Sudden drift occurs when rapid, irreversible changes occur over a short period. Gradual drift occurs when the data distribution gradually replaces old data over time. Recurring drift occurs when the previous data distribution occurs again over time.

#### 1) DRIFT DETECTION

The design and development of a DDoS attack detection model must consider drift detection and adaptation. Drift detection is a crucial component for adaptive DDoS attack detection, as it addresses issues related to concept drift. There are two primary categories of drift detection

methods: performance-based and distribution-based methods [13]. Performance-based drift detection methods employ metrics that evaluate the performance of the detection model, such as increases in error rates and decreases in accuracy. In contrast, distribution-based drift detection methods utilize statistical variables, including mean, variance, and class imbalance.

Performance-based concept drift detection methods rely on changes in the performance metrics used to evaluate DDoS detection models. Two well-known performance-based drift detection methods are the Drift Detection Method (DDM) and the Early Drift Detection Method (EDDM). The DDM is a performance-based approach that utilizes changes in the model's error rate and standard deviation, applying predefined thresholds [14] for detection. DDM often performs better on data streams with sudden drifts. However, the reaction time is often too slow to detect gradual drift. The Early Drift Detection Method (EDDM) is an enhanced version of the DDM that uses the same drift detection mechanism [15]. Although the EDDM frequently performs better than the DDM, it often falls short in data streams with gradual drift.

Distribution-based concept drift detection methods are based on changes in the data distribution measurements of the old and new in a given time window. The change in data distribution often triggers a concept drift to update the learning model. Several statistical variables were used to measure the data distribution over time windows. Mean, variance, and information entropy are commonly used to measure the data distribution. Adaptive windowing (ADWIN) is a widely used method that uses an adaptive sliding window to detect concept drift based on the statistical difference between two adjacent sub-windows. If there is no concept drift, the window size is dynamically increased, whereas it is reduced when a concept drift is detected [16]. IoT Systems with less memory often adopt distribution-based methods because they need only be retained in the most recent samples. Windowing methods are typically quick and simple. However, they may have missed important historical data.

#### 2) DRIFT ADAPTATION

Concept drift must be handled after it is detected by updating the current models using the appropriate drift adaptation methods. Drift adaptation is a procedure used to update a model automatically when concept drift occurs, to enhance the performance of the model and detect zero-day attacks. These procedures usually fully retrain or alter the learning model by using a new dataset. The three categories of drift adaptation methods are model retraining, incremental learning, and ensemble learning.

Model retraining is a simple and straightforward method for handling concept drift. Offline models are often unable to predict unseen incoming streaming data accurately. This problem can be addressed by retraining the model using the most recent data streams. There are two types of model retraining methods: full and partial. Full retraining retrains the learning model using the entire dataset, whereas partial

retraining trains the learning model using selected parts of the dataset. The ADWIN method was used to partially retrain the model by using the most recent data. ADWIN is a drift detection method that uses model retraining. ADWIN performs better because it uses a dynamic window to fit the new data [16]. The learning model was partially retrained on only the new concept samples to reduce the training time.

Incremental learning is commonly used in data-stream analytics. Incremental learning involves updating the learning model when each instance is processed. This method is particularly suited to dynamic environments, in which data streams continuously evolve [17]. They do not require a sufficient amount of data prior to the training process, because of the capacity of incremental learning approaches to support progressive learning. Incremental learning techniques such as SGD, incremental SVMs, and online decision trees enable models to learn incrementally.

Ensemble learning methods have become vital for managing streaming data environments to enable a stronger concept drift adaptation. Ensemble learning methods have been developed to provide powerful base learners for DDoD attack detection and have stronger concept drift adaptation. Ensemble learning uses an averaging algorithm that combines multiple base learners to address the problems [18]. Ensemble learning capitalizes on the diversity of base models to enhance performance because base learners may capture different patterns within the data. The ensemble combines individual predictions from each base learner to make a final prediction by averaging or majority voting during the classification tasks. Using a single model type in an ensemble is significantly more effective for concept drift adaptation than training new models in data streams with recurrent concept drifts [19].

There are two main categories of ensembles used in data stream analytics: online and block-based ensembles [20]. In block-based ensembles, base learner training is performed on fixed-size blocks. Base learners are evaluated and updated each time a new block arrives. Block-based ensembles provide accurate responses to gradual drifts. However, a longer response time was required for a sudden drift. There are three block-based ensembles: accuracy weighted ensemble (AWE) [21], Accuracy Updated Ensemble (AUE) [22], and Streaming Ensemble Algorithm (SEA) [23]. The AUE is designed to maintain an ensemble of base models that are periodically updated with new data without weighting the individual models differently. This simplicity makes the AUE computationally efficient [22]. AUE uses nonlinear error functions to apply weights to base learners in order to improve their performance. Hoeffding trees leverage the Hoeffding bound to create statistically significant splits and allow them to grow incrementally, based on sufficient data. These trees are highly efficient and are suitable for large-scale data streams.

Adaptive Random Forest (ARF) adapts the traditional Random Forest algorithm for streaming data using Hoeffding Trees as base learners and implementing mechanisms to handle concept drift using ADWIN [24]. The ARF includes

a powerful resampling method and flexibility to accommodate various drift types. ARF is highly effective in dynamic environments but demands significant memory and processing resources. The Streaming Ensemble Algorithm (SEA) employs a fixed-size chunking approach to maintain and update an ensemble of base learners incrementally [25]. Each base learner is trained on a specific data chunk. However, a fixed chunk size may not always be optimal for various types of data streams. This makes SEA robust in handling concept drift. Leverage bagging (LB) [26] is another online ensemble that constructs base learners using bootstrap samples. Although LB is simple to construct, it usually performs worse than the SRP and ARF. Although there are several concept drift adaptation strategies in use today, their efficacy is constrained by slow drift response and poor prediction accuracy.

The incremental learning method updates itself as new data arrive without requiring complete retraining from scratch. This method is essential for detecting DoS attacks in continuous data streams. The key characteristics of incremental learning include continuous learning, memory efficiency, and adaptability. Incremental learning methods often perform poorly because of their low model complexity and limited drift adaptability [27]. Block-based ensembles face significant difficulties in determining block size and responding quickly to drift. Online ensembles such as ARF and SRP often outperform incremental learning and block-based ensemble methods. However, owing to their randomization strategies, they provide unstable learning models, adding more unpredictability to DoS attack detection.

## B. ADVERSARIAL ATTACK GENERATION METHODS

The methods used to generate adversarial attacks vary based on the nature of the data and the sector. Below, we describe five adversarial attack generation methods.

### 1) FAST GRADIENT SIGN METHOD (FGSM)

The Fast Gradient Sign Method (FGSM) is an easy and efficient method for generating adversarial perturbation [28]. Adversarial perturbations are inputs that have been intentionally crafted to maximize the amount of perturbation to the input data to misclassify machine learning models. After determining the gradient of the loss function with respect to the input data, the FGSM crafts a perturbation in the direction of the sign of the gradient. The amount of modification of the input data is controlled by a hyperparameter called epsilon ( $\epsilon$ ), which determines the magnitude of perturbation. The FGSM approach must be adapted to operate using online learning algorithms. In online learning, models are updated incrementally as new data points arrive. Therefore, an adversarial attack should be able to perturb data points as they arrive. Given an online learning model  $f(\theta_t, x_t)$  with parameters  $\theta_t$  and data point  $x_t$  at time  $t$  and a loss function  $l(\theta_t, x_t, y_t)$  where  $y_t$  is the true label at time  $t$ , the objective is to create an adversarial example  $x'_t$  that maximizes the loss. For each incoming data point  $x_t$ , generate an adversarial example  $x'_t$



by perturbing  $x_t$  in the direction of the gradient of the loss function with respect to  $x_t$ :

$$x'_t = x_t + \varepsilon \cdot \text{sign}(\nabla_{x_t} l(\theta_t, x_t, y_t)) \quad (1)$$

Perturbation  $\varepsilon$  controls the magnitude of the change in the input, ensuring that the adversarial example remains close to the original input. The constraint is typically applied to the  $l_\infty$  norm:

$$\|x'_t - x_t\|_\infty \leq \varepsilon \quad (2)$$

Compute the gradient of the loss function with respect to the current input  $x_t$ :

$$\nabla_{x_t} l(\theta_t, x_t, y_t) \quad (3)$$

## 2) BASIC ITERATION METHOD (BIM)

The Basic Iteration Method (BIM) first proposed by Kurakin et al. [29] in 2017. BIM is a basic extension of the FGSM. BIM adopts FGSM in an iterative approach, rather than applying multiple inputs with small step-size perturbations in the direction that maximizes the loss of the model. The intention is to provide an adversarial example that can deceive the model's predictions while still seemingly comparable to the original input. BIM generates adversarial examples by iteratively applying FGSM with a smaller step size. The details of the mathematical formulation of BIM, particularly in the context of online learning, are formulated as follows:

For each iteration  $i$ , initialize:

$$x_{adv}^{(0)} = x_t \quad (4)$$

Iterative update

$$x_{adv,t}^{(i+1)} = x_{adv,t}^{(i)} + \alpha \cdot \text{sign}(\nabla_{x_{adv,t}^{(i)}} \mathcal{L}(x_{adv,t}^{(i)}, y_t; \theta_t)) \quad (5)$$

where,  $\alpha$  is the step size and  $(\nabla_{x_{adv,t}^{(i)}} \mathcal{L}(x_{adv,t}^{(i)}, y_t; \theta_t))$  is the gradient of the loss with respect to the adversarial example at iteration  $i$ .

Clip perturbation:

$$x_{adv,t}^{(i+1)} = \text{clip}(x_{adv,t}^{(i+1)}, x_t - \epsilon, x_t + \epsilon) \quad (6)$$

where  $\epsilon$  is the maximum perturbation allowed and  $\text{clip}(\cdot)$  ensures that the perturbation remains within the range  $[-\epsilon, +\epsilon]$ .

After  $N$  iterations, the final adversarial example is:

$$x_{adv,t=N} = x_{adv,t}^{(N)} \quad (7)$$

## 3) PROJECTED GRADIENT DESCENT (PGD)

Projected Gradient Descent (PGD) is an iterative optimization algorithm [30], [31]. This is designed to minimize the loss function. PGD builds upon the principles of FGSM but introduces an iterative refinement process to overcome some of FGSM's limitations. The PGD operates by iteratively applying small perturbations to the input in a manner that maximizes the loss function of the model while remaining

within a specified perturbation budget. This iterative nature allows the PGD to explore a wider range of perturbations. Mathematically, is represented as

$$\theta_{t+1} = P_c(\theta_t - \alpha_t \nabla f(\theta_t, x_t, y_t)) \quad (8)$$

where,  $x_t$  is the input at iteration  $t$ ,  $\alpha_t$  is the step size,  $\nabla f(\theta_t, x_t, y_t)$  is the gradient of the loss function with respect to the input and  $P_c(\cdot)$  is projection operator ensuring perturbed input stays within predefined bounds.

## 4) CARLINI AND WAGNER (C&W)

A C&W attack is a state-of-the-art optimization-based attack [31]. CW is a highly effective adversarial attack method that can be adapted for online learning scenarios to generate minimal perturbations that mislead machine learning models. There are two variants of CW attack: L2 and L $\infty$ . The CW objective is to minimize the following:

$$\min_{\delta} \|\delta\|_2 + c \cdot f(x_t + \delta) \quad (9)$$

where  $\|\delta\|_2$  is the  $\mathcal{L}_2$  norm of the perturbation, and  $c$  is a constant balancing the trade-off. The function  $f(x_t + \delta)$  defined as:

$$f(x_t + \delta) = \max(0, Z(x_t + \delta)_y, \max_{i \neq y} -Z(x_t + \delta)_i + \kappa) \quad (10)$$

where  $Z(x_t + \delta)$  denotes the logits of the model,  $Z(x_t + \delta)_y$  is the logit for the true class,  $Z(x_t + \delta)_i$  is the logit for the target class, and  $\kappa$  is a confidence parameter.

## 5) GENERATIVE ADVERSARIAL NETWORKS (GANS)

A Generative Adversarial Network (GAN) is a deep learning architecture that competes with two neural networks against one another in the context of a zero-sum game. The GAN was initially introduced by Goodfellow to estimate the generative models [32]. GANs aim to produce new synthetic data that resemble well-known data distributions. Generative Adversarial Networks (GANs) are neural networks used for unsupervised learning. The generator created fake data samples to deceive the discriminator. In contrast, the discriminator attempts to distinguish between real and fake samples. Both the generator and discriminator are neural networks, and they compete with each other during the training phase. The procedures are repeated several times, and the generator and discriminator become better at what they are doing. GANs have diverse applications, such as producing synthetic data to increase the number of data samples and resolving under-sampling issues [33]. A newly emerging area of research focuses on integrating machine-learning technology with privacy-protection technology.

The training process in online learning GANs is formulated as a minimax game, with the objective function  $V(G, D)$  expressed as:

$$\begin{aligned} \min_G \max_G V(D, G) \\ = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \end{aligned} \quad (11)$$

In each iteration, the discriminator  $D$  aims to maximize its ability to differentiate between real samples  $x$  from the current data batch and fake samples  $G(z)$ , where  $z$  is a noise vector.

$$\max_G \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \quad (12)$$

Simultaneously, the generator  $G$  attempts to minimize the discriminator's ability to correctly identify its synthetic samples as fake.

$$\min_G \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] \quad (13)$$

This iterative adversarial process continues as new data arrives and ensures that both networks improve over time.

### C. ADVERSARIAL ATTACK AND DEFENSE IN IOT SYSTEM

In this section, we explore the most recent literature on adversarial attacks and defense in the IoT network context. Specifically, we focus our review on recent research concerning adversarial attacks and defense against DDoS attacks. This review provides an overview of the models, techniques, and tools utilized for adversarial attacks and defense, as well as the datasets used to evaluate these systems. Zhou et al. [34] proposed a state-of-the-art method for generating adversarial attacks, called hierarchical adversarial attacks (HAA). The objective of this method is to operate within a specified budget limit and construct a sophisticated level-aware black-box attack against a GNN-based IDS in IoT networks. The authors evaluated their HAA method using the UNSW-SOSR2019 dataset. The experimental results show that adversarial attacks based on the HAA method reduce the classification precision of both GNN models by more than 30%. Kotak and Elovici [35] proposed a new method to generate a real-time adversarial example using heatmaps from Grad-CAM++ and Class Activation Mapping (CAM). The authors explored the vulnerability of fully connected neural networks (FCN), convolutional neural networks (CNN), and Global Average Pooling (GAP) using the IoT Trace dataset. Surprisingly, adversarial examples can be transferred to different model architectures. The GAP model behaves differently, suggesting that it may be a robust model for adversarial examples.

Yumlembam et al. [12] proposed a new approach called VGAE-MalGAN, which can effectively add nodes and edges to an existing API graph and dynamically generate an adversarial network against GNN-based malware detection. The authors developed robust Android malware detection using VGAE-MalGAN adversarial training and achieved a high malware detection accuracy. Ibitoye et al. [36] demonstrated the effects of adversarial samples on a deep learning-based Intrusion Detection System (IDS) using the BoT-IoT dataset. They conducted a performance comparison of two neural networks: a feedforward neural network (FNN) and a Self-normalizing Neural Network (SNN), employing metrics

such as accuracy, precision, and recall. The authors used the Fast Gradient Sign Method (FGSM), the Basic Iteration Method (BIM), and Projected Gradient Descent to craft adversarial samples. The experimental results show that the self-normalizing feature enhance the SNN more resilient to gradient-based adversarial samples, while the FNN outperforms the SNN in intrusion detection within IoT networks.

Papadopoulos et al. [37] evaluated traditional machine learning and deep learning models using Bot-IoT datasets. The attacker generates adversarial examples employing the Fast Gradient Sign Method (FGSM) to evade detection measures against binary and multiclass Artificial Neural Networks (ANNs). Additionally, the author analyzed and conducted various experiments to demonstrate how the attacker could circumvent the detection model with significant probability, highlighting the robustness of the detection model against adversarial examples. Since the attacker possesses knowledge of datasets and models, this scenario is classified as white-box attack. Experimental results indicate that an attacker can manipulate or circumvent detection with a significant probability. Benaddi et al. [38] proposed an adversarial training method to improve the effectiveness of a hybrid CNNLSTM-based Intrusion Detection System (IDS) using Conditional Generative Adversarial Networks (C-GAN). The authors integrated C-GAN into the training pipeline to address classes with small sample sizes and to mitigate data imbalance in the BoT-IoT dataset. Initially, the BoT-IoT dataset was utilized to train the IDS model, and certain low-performing classes were identified. These classes were then used to train the C-GAN, and the C-GAN generator was used to retrain the IDS model, which enhanced the performance of the identified classes.

Benaddi et al. [39] employed a similar approach to improve the robustness and efficacy of an IDS in the IIoT. The study suggests employing Deep Reinforcement Learning (DRL) in conjunction with GAN to improve the performance of an IDS. Experiments were conducted on the Distributed Smart Space Orchestration System (DS2OS) dataset, and it was found that the proposed DRL-GAN model performs better than the standard DRL in identifying anomalies in imbalanced datasets related to the IIoT. However, during the training phase, the proposed approach required a significant amount of computational resources. Jiang et al. [40] proposed a novel defense framework for an IDS against adversarial attacks on an IoT network. This framework is known as a feature grouping and multi-model fusion detector (FGMD). This framework integrates different models, each of which processes distinct portions of input data or features to better withstand the effects of adversarial attacks. The authors used two publicly available datasets: MedBioT and IoTID to validate the proposed model using adversarial examples. They utilized DT, LSTM, and RNN as the baseline models for comparison. The experimental results demonstrated the effectiveness of FGMD against adversarial attacks. FGMD outperform the baseline models in terms of detection rate.

**TABLE 1.** Summary of related works.

| Author & year            | Model                                                                   | Attack technique                                | Defence  | Learning method        | Dataset                        |
|--------------------------|-------------------------------------------------------------------------|-------------------------------------------------|----------|------------------------|--------------------------------|
| [35](2022)               | GNN                                                                     | HAA                                             | X        | Batch learning         | UNSW-SOSR2019                  |
| [36](2023)               | CNN, GAP and CNF                                                        | Grad-CAM++ nad CAM                              | X        | Batch learning         | IoT Trace dataset              |
| [12](2023)               | GNN                                                                     | VGAE-MaIGAN                                     | X        | Batch learning         | CICMaldroid and Drebin dataset |
| [37](2020)               | FNN & SNN                                                               | FGSM & BIM                                      | X        | Batch learning         | BoT-IoT dataset                |
| [38](2021)               | ANN and SVM                                                             | FGSM                                            | X        | Batch learning         | Bot-IoT datasets               |
| [39](2022)               | DRL and GAN                                                             | X                                               | X        | Batch learning         | DS2OS dataset                  |
| [41](2022)               | DT, LSTM, and RNN                                                       | FGMD                                            | X        | Batch learning         | MedBioT and IoTID              |
| [42](2025)               | TTOS-RAAM                                                               | X                                               | CVAE     | Batch learning         | RT-IoT2022 dataset             |
| [43](2024)               | RF, KNN, XGB and Voting                                                 | X                                               | X        | Batch learning         | MTA-KDD'19                     |
| [44](2023)               | LSTM-CNN                                                                | FGSM, BIM, MIM, PGD, JSMA, Boundary Attack, C&W | ✓        | Batch learning         | LwHBench dataset               |
| [45](2024)               | CNN-RF                                                                  | X                                               | X        | Batch learning         |                                |
| <b>Proposed Solution</b> | <b>ARF-ADWIN, ARF-DDM, HT, KNN-ADWIN, SRP-DDM, SRP-ADWIN, and PWPAE</b> | <b>FGSM, BIM, MIM, PGD, C&amp;W and GAN</b>     | <b>✓</b> | <b>Online learning</b> | <b>IOTID20 and CICIoT2023</b>  |

Prasad et al. [41] proposed the Two-Tier Optimization Strategy for Robust Adversarial Attack Mitigation (TTOS-RAAM) model for IoT intrusion detection system. This hybrid framework combining Conditional Variational Autoencoders (CVAE) for adversarial detection and ICAVO for parameter tuning to address adversarial attacks. The attack type involved adversarial samples targeting IoT network traffic. Defense methods included feature optimization with CGWO and parameter tuning with ICAVO. The model achieved 99.91% accuracy on the RT-IoT2022 dataset; However, it lacked adaptive attack prevention mechanisms?. Ramaiah et al. [42] proposed an ensemble classifier utilizing the Optimal Correlation Strategy (OCS) for feature selection in malware detection systems. While the defense method employed a robust feature selection approach it didn't incorporate defense against adversarial attack. The model demonstrated strong performance on the MTA-KDD'19 malware datasets, but its adaptability to zero-day attacks was limited.

Sánchez et al. [43] employed an LSTM-CNN architecture based on hardware performance behavior for individual device identification. They conducted adversarial attack using deep learning evasion attacks. The adversarial defense strategies included adversarial training and the model distillation method to improve robustness. The model's robustness increased from an attack success ratio of 0.88 to 0.96. The experiment performed using LwHBench dataset?. Ma et al. [44] proposed a method for detecting Low-rate Denial of Service (LDoS) attacks in Artificial Intelligence of Things (AIoT) environments. The authors introduced a novel cooperative defense scheme against hybrid LDoS attacks. They utilized programmable utilized the plane of Software-Defined Networking (SDN) to synchronize

real-time and high-precision defenses. Experimental results of LDoS demonstrated that their method detects more accurately and mitigates effectively than the traditional method.

The reviewed research papers primarily focus on batch learning methods for DDoS detection, neglecting the dynamic nature of IoT environments. Consequently, these approaches fail to address concept drift, which occurs when data distributions change over time. None of the examined papers propose an adversarial defense for online DDoS detection systems. Furthermore, they are unable to manage multiple and unknown adversarial attacks in real-time environments. In this study, we propose a multi-armed bandit (MAB) approach with Thompson Sampling (MLBTS) for an online DDoS attack detection system that effectively addresses these challenges.

### III. PROPOSED FRAMEWORK

#### A. OVERVIEW OF PROPOSED FRAMEWORK

The proposed framework is a comprehensive solution designed to protect online DDoS attacks detection systems against adversarial attacks in IoT environments. It integrates multiple components to achieve a robust online DDoS detection system. The framework consists two main components: pre-processing and the Multi-Stage Adversarial Attack Defense (MSAAD) framework. The MSAAD framework is designed to protect IoT systems from DDoS attacks through a combination of Resilient Adversarial Detection and Purification (RADP), multiple classifiers, and ensemble defense strategies. These components interact seamlessly to ensure real-time, robust performance against multiple and unknown adversarial attacks.

### 1) PRE-PROCESSING

Pre-processing ensures clean, well-represented, and balanced IoT data streams for optimal model DDoS attack detection accuracy. This step includes encoding, cleaning, sampling, balancing, normalization, and feature selection of data. These techniques prepare IoT data streams for online DDoS attack detection.

### 2) MULTI-STAGE ADVERSARIAL DEFENSE (MSAAD)

The proposed MSAAD composed of three layers of defense:

- *RADP (First Defense Layer)*

The Resilient Adversarial Detection and Purification (RADP) designed to detects and purifies adversarial attacks against online DDoS attack detection systems addressing from multiple and unknown adversarial attacks. The RADP consists two main components: Resilient Adversarial Detector (RAD) and purification modules. The RAD module detects adversarial attacks by leveraging the strengths of Generative Adversarial Networks (GANs) and multisource adversarial perturbations (MSAP). A GAN are employed to reduce the gap between the training data and the real-world data. MSAP crafts a variety of adversarial attack methods to create diverse perturbations to train adversarial attack detectors. The purification module is used to purify adversarial perturbations to enhance the robustness of online learning models by systematically removing the detected data points using RAD.

- *Multiple Classifiers (Second Defense Layer)*

This layer consists of multiple classifiers designed to defend against adversarial attacks. Utilizing multiple classifiers significantly makes it harder for attackers to exploit the system. Each classifier is optimized for different traffic patterns, ensuring high accuracy and robustness against sophisticated attack strategies.

- *Ensemble Defense (Third Defense Layer)*

The third layer of defense is the Multi-Armed Bandit with Thompson Sampling (MABTSE), which operates by balancing exploration (trying different online learning models) and exploitation (using the best-known online learning model) to dynamically select an optimal classifier or ensemble for each data point. For the ensemble, we employed the Accuracy Update Weighted Probability Averaging Ensemble (AUWPAE) approach to combine the base learners (classifiers) for IoT data stream analytics [11].

### 3) DATA PRE-PROCESSING

Data preprocessing is essential to ensure that incoming data are clean, well-represented, and balanced for optimal model performance. The performance of the DDoS attack detection method can be significantly affected by the representation, size, and quality of incoming data. In particular, selecting a dataset with high dimensionality and a large number of duplicate and irrelevant features can adversely impact the training

process. To address these issues, we employed data cleaning, encoding, normalization, and feature selection techniques.

### 4) ENCODING

IoT data values were generated as words and strings to enhance readability. Data encoding involves converting string information into numerical features that can be understood and processed by online learning models. One-hot encoding and target encoding are examples of frequently used data encoding methods. Hot encoding is a common method used to represent each category as a binary vector. However, in an online environment, incremental one-hot encoding is used to dynamically add new categories to the encoding matrix as data arrive. In this study, we utilized incremental one-hot encoding to handle the dynamic nature of the IoT data.

### 5) DATA CLEANING

IoT streams often experience missing values due to data accessibility issues. These gaps are typically addressed using incremental imputation techniques, such as mean or median imputation or last observation carried forward (LOCF), as data arrives. In an IoT environment, data distribution can change over time. Therefore, the cleaning process must be adaptive, and its parameters should be continuously updated to reflect the current state of the incoming data stream. In this study, we employed the missing values using mean or median imputation techniques.

### 6) SAMPLING

The selection of highly representative data samples necessitates an efficient data-sampling method. In online learning, sampling is conducted using incremental K-means and mini-batch K-means. In incremental K-means clustering, data points are assigned to the closest cluster centroid. The centroid is updated incrementally when new data arrives. This approach enables the online DDoS detection model to dynamically maintain its cluster, without requiring access to the entire dataset. Mini-batch K-means is other method used to process small batches of data simultaneously. This makes it more scalable and efficient for an online DDoS attack detection model. The proposed framework employs a mini-batch K-mean-based cluster sampling method to obtain highly representative data.

### 7) DATA BALANCING

The sampling method is effective in handling class imbalances in machine learning. The Synthetic Minority Oversampling Technique (SMOTE) generates synthetic samples by adding existing minority class samples to create a balanced dataset. We use the online SMOTE sampling method to generate synthetic samples when new minority class instances arrive. This helps to prevent our framework from becoming biased towards the majority class. This leads to improved performance of the proposed framework.



## 8) DATA NORMALISATION

Data normalization is a key pre-processing step in online learning to ensure that the feature values are within the same range. In online learning, data arrive incrementally; therefore, normalization must be performed dynamically. Two common normalization methods used on data analytics are the Z-score and min-max normalization. The min-max normalization approach transforms the original data linearly to achieve a balance of comparative values between the data before and after processing. The min-max normalization method scales the values of a feature to a range between 0 and 1. The Min-Max scaling approach is shown in Equation 14. [45]

$$X_{new} = \frac{X - \text{Min}(X)}{\text{Max}(X) - \text{Min}(X)} \quad (14)$$

where  $X_{new}$  is the result of the normalization,  $X$  is the value to be normalized, and Min and Max are the minimum and maximum values of each feature, respectively.

The Z-score normalization method was based on the mean and standard deviation of the data. This method is useful if the minimum and maximum data values are unknown. Equation 15 shows the formula for Z-score normalization [46].

$$X_{new} = \frac{X - \mu}{\sigma} \quad (15)$$

where  $X_{new}$  is the result of normalization,  $X$  is the value to be normalized,  $\mu$  is the population mean, and  $\sigma$  is the standard deviation. min-max normalization retains outliers in the datasets, making it appropriate for DDoS attack detection models. Therefore, the proposed framework uses min-max normalization to address this requirement.

## 9) FEATURE SELECTION

Feature selection is employed to improve the online learning model performance by reducing noise and focusing on the most relevant features. The optimal minimal feature set can be determined by evaluating all possible feature combinations. However, when dealing with the large number of features, optimization approaches can be utilized to explore the feature search space and determine the optimal feature set. Information Gain (IG) and Pearson Correlation methods were utilized to remove irrelevant and duplicate features, respectively. Information Gain (IG) is used to remove irrelevant features by measuring the amount of information that contributes to predicting the target variable. Pearson Correlation helps identify and remove duplicate features by evaluating the linear relationships between pairs of features. Consequently, the proposed framework selects both Information Gain and Pearson Correlation to identify the best-performing minimal feature set.

## B. MULTI-STAGE ADVERSARIAL ATTACK DEFENCE (MSAAD) FRAMEWORK

The proposed MSAAD framework operates through three key stages: Resilient Adversarial Detection and Purification (RADP), multiple classifiers, and ensemble defense.

The RADP defense method detects and purifies adversarial attacks against an online DDoS attack-detection system, addressing multiple and unknown adversarial attacks. RADP consists of Resilient Adversarial Detection (RAD) and purification modules. RAD is trained using a generative adversarial network (GAN), that mimic crafted adversarial samples and multi-source adversarial perturbation generator (MSAPG) that generates adversarial perturbation, as shown in Figure 1. The GAN leverage its strengths to reduce the gap between the training data and real-world data. This makes the online DDoS detector more aware of multiple and unknown adversarial attacks. MSAPG generates various adversarial attack methods to create diverse perturbations to train an adversarial attack detector. It includes adversarial training by crafting perturbations using PGD, FGSM, C&C, and BIM methods. This ensures that the adversarial detector is trained on diverse adversarial perturbations to make it more robust and resilient to a broad spectrum of attacks. By combining the GAN and MSAPG submodules, the proposed MSAAD framework significantly enhances the detection capability of the RAD. The Pruning method is used to purify adversarial perturbations to enhance the robustness of online learning models by systematically removing detected data points using RAD. The details of the MSAPG framework are shown in Figure 1.

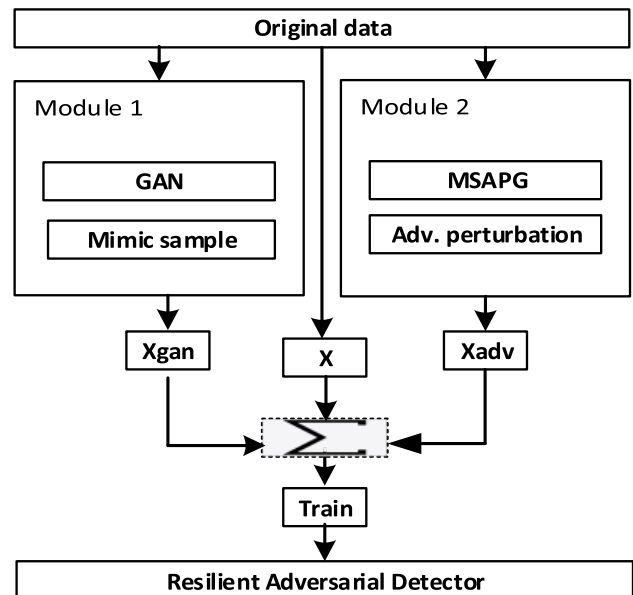


FIGURE 1. Training resilient adversarial detection model through MSAPG.

The second stage of MSAAD employs a multiple-classifier approach to further protect the DDoS attack detection system against adversarial attacks. The concept behind utilizing multiple classifiers as a defense instead of using a single classifier is to increase the complexity of attackers when attempting to replicate the DDoS attack detection model. If an attacker manages one classifier to bypass it, it is difficult to fool and bypass multiple classifiers. An attacker cannot predict the classifier responsible for classifying a given stream of

traffic. This is because they cannot predict the specific model responsible for classifying a given request. This enhances the overall resilience of DDoS attack detection systems to sophisticated adversarial attacks.

The third stage of MSAAD employs the Multi-Armed Bandits (MAB) algorithm with Thompson Sampling (MABTSE) to dynamically select the optimal classifier or ensemble of classifiers for each incoming traffic request. For the ensemble, the Accuracy Update Weighted Probability Averaging Ensemble (AUWPAE) approach is employed to combine the base learners (classifiers) for IoT data stream analytics. This allows for seamless integration and enhances adaptability. It maintains the performance of conventional DDoS attack detection systems in non-adversarial attack scenarios by selecting the most appropriate classifiers for each request type. The MAB algorithm was specifically designed to account for the diversity of classifiers within the system. For example, if the SRP-ADWIN classifier demonstrates a high probability of being accurate, whereas the SRP-DDM and KNN-ADWIN classifiers have lower probabilities, the MAB algorithm will prioritize the SRP-ADWIN for that particular request. This process optimizes the selection of classifiers and enhances the overall accuracy of the DDoS attack-detection system. Moreover, the MAB algorithm continually updates the probability estimates for each classifier based on their recent performance, and allows the system to adjust traffic patterns and stay up-to-date with emerging types of attacks.

The MAB algorithm with Thompson Sampling dynamically selects an optimal classifier or an ensemble of classifiers for each incoming traffic request. The DDoS attack detection system balances exploration and exploitation by selecting classifiers that have achieved recent success but are adaptable to multiple and unknown types of attacks. In some cases, multiple classifiers exhibit similarly high probabilities of success. To handle this situation, an ensemble method was applied to combine the outputs of the classifiers. In this study, we used the Accuracy Update Weighted Probability Averaging Ensemble (AUWPAE) approach to combine the base learners for IoT data-stream analytics. Details of this algorithm are presented in Algorithm 1. AUWPAE provides dynamic weights to base learners in real time. The AUWPAE method uses a weighted average approach. The prediction probability of each base learner is assigned a weight and multiplied by the prediction probability of that base learner. The multiplication results are summed to determine the final prediction probability. AUWPAE mitigates the limitations of other online ensemble models. It addresses sudden and gradual drift issues and enhances the DDoS attack detection. ARF-ADWIN, ARF-DDM, SRP-ADWIN, SRP-DDM, and KNN-ADWIN were used for ensemble-based learners.

## IV. EXPERIMENT

### A. DATASET

We utilized two security datasets, CICIoT2023 and IoTID20, to evaluate the proposed online MSAAD framework.

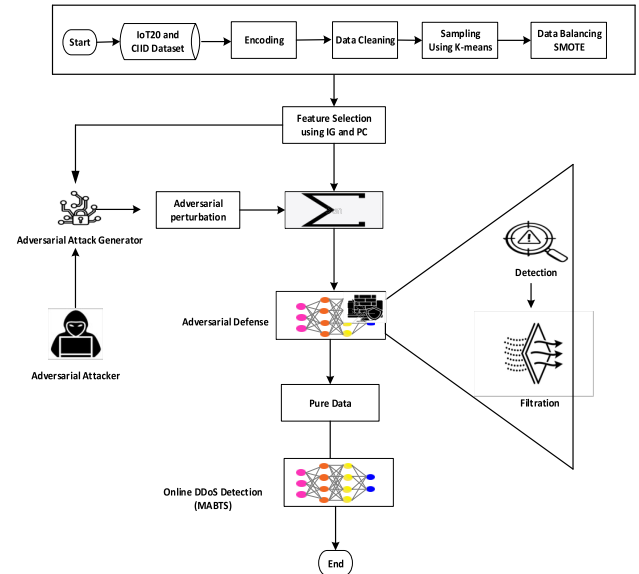


FIGURE 2. Overview of MSAAD framework.

The Canadian Institute for Cybersecurity developed the CICIoT2023 dataset by 2023 [47]. Its objective was to provide a novel and extensive dataset for research and development in the IoT cybersecurity domain. To generate attack traffic, 33 attacks were executed on 105 IoT devices. Traffic includes both normal and attacked traffic. Attack traffic is classified into seven categories: DDoS, DoS, recon, web-based DDoS, brute-force, spoofing, and mirai. This dataset comprises both the common and current attacks. However, in this experiment, we used the DDoS and normal traffic.

The IoTID20 dataset has been utilized to develop DDoS attack-detection methods in several studies [48]. The authors of the IoTID20 dataset developed different binary and multi-class classifiers and evaluated the accuracy using the IoTID20 dataset [49]. The attack traffic for the Mirai botnet was generated separately using a laptop and was later changed to simulate its origin from IoT devices. DoS, scanning, and man-in-the-middle attacks were simulated using the NMap tools. DDoS attack types, such as mirai ACK flooding, mirai brute force, mirai HTTP Flooding, and mirai UDP Flooding, were included. IoTID20 was selected because it is publicly available, and several recent studies have used it to develop IDS [50]. IoTID20 focuses on IoT security, and provides a wide range of attacks and normal samples from various IoT devices.

### B. EXPERIMENT ENVIRONMENT

The proposed online DDoS detection framework is designed to detect DDoS attacks targeting IoT systems. To evaluate the performance of the proposed approach, we developed a prototype using the Python 3.10 within the Jupyter Notebook environment. The River library [51] is utilized for data stream analytics, and addresses concept drift through machine learning. The Adversarial Robustness Toolbox (ART) [52]

**Algorithm 1** MAB With Thompson Sampling and AUWPAE (MBATSE)**Input:**  $X_{train}, y_{train}, X_{test}, y_{test}$ **Output:**  $t, m$  (time steps and accuracy metrics)

```

1 function TS-WPAE ( $X_{train}, y_{train}, X_{test}, y_{test}$ )
2   Initialize  $a, t, e, eps = 0.0001$ 
3   Initialize base learners  $j = 5$ 
4   Thompson Sampling parameters  $\alpha = 1, \beta = 1$ 
5   for each instance  $(x_i, y_i)$  in  $X_{train}, y_{train}$  do . //
      learn base learners using training set
6      $L_j \leftarrow (x_i, y_i)$ 
7   end for
8   for each instance  $(x_i, y_i)$  in  $X_{test}, y_{test}$  do . //
      predict using test set
9      $\theta_j \leftarrow \text{sample}(j, j)$ 
10     $\theta_{\text{theta\_value\_top\_three\_baselearner}}(\theta_j)$ 
11     $y_{pred_j} \leftarrow \text{bl}(x_i)$  //predict target
12     $y_{prob_j} \leftarrow \text{blProbPred}(x_i)$  //target prediction
      probability
13     $\text{bl.update}(x_i, y_i)$  //update base learner.
14  end for
15  for  $j = 1$  to  $3$  do
16     $e_i \leftarrow \text{MSE}_i(y_i, y_{pred_j})$  // calculate real-time
      MSE for each base learner
17     $W \leftarrow 1/(e_j + eps)$  //calculate weight
18     $y_{prob_{j0}} \leftarrow y_{prob_j}$  // probability for class 0
19     $y_{prob_{j1}} \leftarrow y_{prob_j}$  //probability for class 1
20     $y_{prob_0} \leftarrow y_{prob_j} \sum_{j=1}^4 W_j * y_{prob_{j0}}$  //average
      weighted probability for class 0
21     $y_{prob_1} \leftarrow y_{prob_j} \sum_{j=1}^4 W_j * y_{prob_{j1}}$  //average
      weighted probability for class 0
22  end for
23  If  $y_{prob_0} > y_{prob_1}$ 
24     $y_{pred} \leftarrow 0$ 
25  else  $y_{prob_0} < y_{prob_1}$ 
26     $y_{pred} \leftarrow 1$ 
27  calculate accuracy, precision, recall and ,
      F1-score;
28  return  $t, a$ 
29 end function

```

employed to simulate adversarial attacks. The experiment was conducted on a machine running equipped with Intel(R) Xeon(R) CPU @2.20GHz with 16 GB of RAM.

**C. EXPERIMENT SETUP**

In this experiment, we utilized state-of-the-art online adaptive learning models as baseline to evaluate the performance of the proposed MABTSE model. ARF-ADWIN, ARF-DDM, SRPs-ADWIN, SRPs-DDM, KNN-ADWIN, HTs, LB, and PWPAE are the baseline models. We utilized four common types of adversarial attacks including PGD, FGSM, C&C, and BIM to generate adversarial perturbations to evaluate MABTSE framework. We have done hyperparameter

tuning carefully to optimize the performance of the proposed online learning model. The ARF tuned with 3 trees and utilized ADWIN and DDM drift detectors to handle concept drift effectively. The KNN model utilizes 5 neighbours and ADWIN concept drift detector using windows size of 100 samples. The SRP and LB ensembles were configured with three base learners. The OPA algorithm was set with an aggressiveness parameter of 1.0. The SAM-KNN and EFDT were fine-tuned to handle non-stationary environments by leverage incremental updates. The generation of adversarial attack perturbation conducted using default parameters, specifically epsilon values of 0.1 to simulate realistic attack scenarios. These hyperparameter configurations were applied in the experiment to achieve optimal performance multiple and adversarial scenarios.

We evaluate the framework against adversarial attacks using three scenarios. In the first scenario, we evaluated the proposed framework as a baseline without adversarial attacks or the RADP defense method. In the second scenario, we conduct adversarial attacks using FGSM, BIG, PGD, and C&W, against the MBATSE framework. The third scenario was designed to evaluate the proposed MSAAD framework, which is mainly utilizes RADP, to detect and purify detected adversarial attacks. This demonstrate the necessity and effectiveness of this defense framework in mitigating adversarial attacks and maintaining the reliability of online DDoS attack detection in IoT systems. The following procedures were used to evaluate the proposed defense framework:

- 1) First, we evaluate the proposed online DDoS detection model, MABTSE, using the original dataset. This serves as the baseline without adversarial attack.
- 2) Generate adversarial perturbation using Generative Adversarial Networks (GANs), and multisource adversarial perturbation generator (MSAPG) submodules and combined with the original training dataset. The method simulates a disrupted input dataset for adversarial attacks.
- 3) The performance of the online DDoS attack-detection model for adversarial samples generated in Procedure 2 was evaluated. This demonstrates the performance of the model under adversarial attack.
- 4) Develop an adversarial sample detection model utilizing the same online model. This model is trained using a combination of the original training dataset and adversarial samples to effectively distinguish between adversarial samples and the original dataset.
- 5) The detected adversarial perturbation was purified from the disrupted input dataset by filtering the detected adversarial perturbations.
- 6) Evaluate the performance of the model to demonstrate the effectiveness of MSAAD framework.

The performance of the proposed online DDoS detection model (MABTSE) and MSAAD defense framework was evaluated by comparing their accuracy, precision, recall, and F1-score, learning time, latency, CI and p-value, as described in Section V. The evaluation utilized CICIOT2023 and

IoTID20 datasets. The evaluation results of the online DDoS attack-detection model are presented in three stages for comparison: original, under attack, and after recovery. These stages are described in detail in Section IV-B, procedure 1,3 and 6. The experiment began with the evaluation of online DDoS attack detection with the original dataset. Following this, the online DDoS attack detection model was evaluated under PGD, FGSM, C&C, and BIM attacks. Subsequently, the Resilient Adversarial Detection and Purification (RADP) method was utilized to detect and purify the detected adversarial attack. Finally, we retrained and evaluated the performance of the online DDoS attack-detection model after recovery. The effects of adversarial attacks against online DDoS attack detection, utilizing the original datasets during and after the defense mechanism are presented in Table 1 and Figure 3 and 4 based on IoTID20 and CICIOT2023 datasets.

#### D. PERFORMANCE METRICS

The proposed MSAAD framework was analyzed from various perspectives to provide a holistic view of the experiment. Since the proposed framework utilized an online DDoS detection model, it reduces memory requirements for data storage and learn new data patterns by processing one data sample at a time. The framework's performance was evaluated using the accuracy, precision, recall, and F1-score metrics. These metrics are commonly employed in machine learning research to provide a well-rounded view of model performance. Due to the inherent imbalance in intrusion detection datasets, relying solely on individual performance metrics such as accuracy, precision, or recall can lead to limited understanding of model performance. Therefore, we considered the accuracy, precision, recall, and F1-scores collectively to ensure a comprehensive comparison of the machine learning models and to avoid skewed evaluation results. Additionally, we include learning time, latency, confidence interval and p-value for the accuracy metrics.

The learning time and latency are utilized to evaluate the proposed framework in real-time scenario. The learning time of the proposed online model, along with others, are analyzed to compare the learning efficiency. This analysis provides insight into the processing time and efficiency requirements of network systems. latency refers the delay in the response time to a specific input stream. It is defined as the processing time required our model to analyze and classify each sample. Low latency an essential performance requirement for online learning to analyze streaming data. Efficient learning model should balance between prediction accuracy and latency to achieve real-time analytics. In the context of network DDoS attack detection and prevention solutions, it is crucial to strike a balance between effectiveness and efficiency should to ensure optimal performance while minimizing computational overhead.

#### V. RESULTS DISCUSSION

Adversarial attacks exploit the vulnerabilities of online learning models by generating adversarial perturbations that can

deceive or manipulate models to make incorrect predictions [52]. Such attacks pose a significant threat to IoT systems and their reliabilities. This experiment aims to demonstrate the detrimental effect of adversarial attacks on the proposed online learning model. Additionally, the proposed defense strategies designed to mitigate these attacks and enhance the performance of DDoS attack detection in IoT systems.

#### A. MABTSE MODEL USING ORIGINAL DATASET

The first result demonstrates the performance of online DDoS attack detection model, known as MABTSE, utilizing the original dataset. This model serves as the baseline for comparison. The performance of the proposed online DDoS attack detection model was compared with other state-of-the-art online adaptive learning methods, including ARF-ADWIN, ARF-DDM, KNN-ADWIN, SRP-DDM, SRP-ADWIN, HT, and PWPAAE. The experiment employed IOTID20 and CICIOT2023 datasets. The attack patterns within both datasets evolved over time, leading to concept drifts of three and seven on the IOTID20 and CICIOT2023 datasets, respectively. These concept drifts are illustrated in Figures 3 and 4 indicated by black arrows. The experimental results are presented in Figures 3 and 4, and Tables 1 and 2.

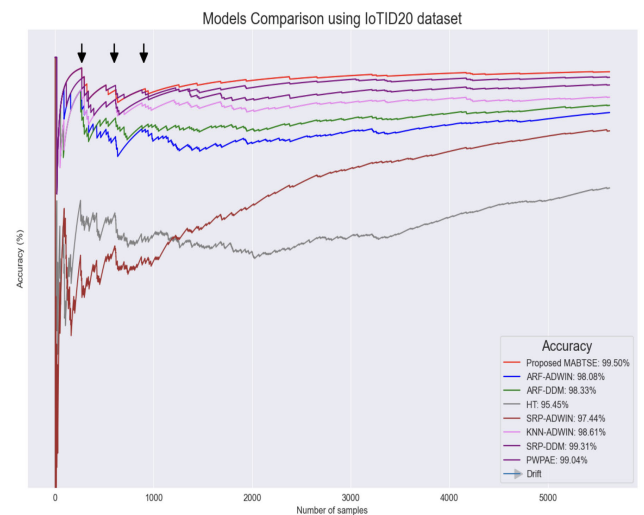


FIGURE 3. Model comparison using IoTID20 dataset.

Figure 3 and Table 1 show the performance comparison of the proposed MABTSE model with other online adaptive models, including ARF-ADWIN, ARF-DDM, SRP-ADWIN, SRP-DDM, HT, KNN-ADWIN, and PWPAAE, utilizing the IOTID20 dataset. The performance metrics employed for this comparison were accuracy, precision, recall, F1-score, and learning time, latency, CI and P-value. The experimental results indicate that the proposed model achieved an average accuracy of 99.50%. This result demonstrates that MABTSE outperforms other online adaptive learning methods. The reason why the proposed model outperforms the others is mainly attributed to the Multi-Armed Bandits (MAB)



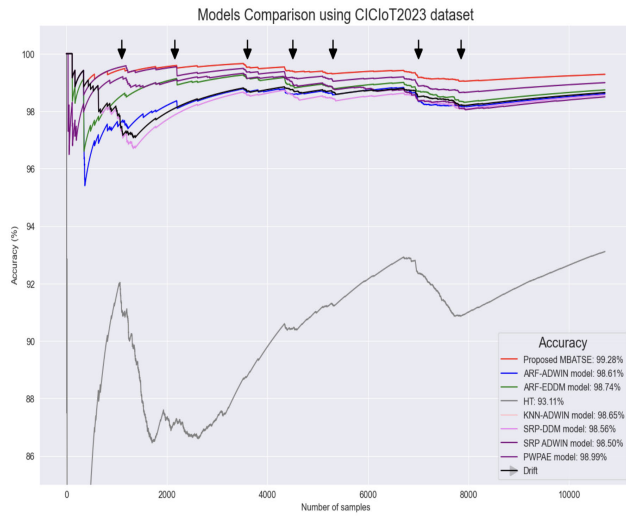


FIGURE 4. Models comparison using CICIDS2017 dataset.

TABLE 2. Proposed online DDoS attack detection model comparison using the IoTID20.

| Models comparison using IoTID20 dataset |              |              |            |              |             |            |
|-----------------------------------------|--------------|--------------|------------|--------------|-------------|------------|
|                                         | A            | P            | R          | F1           | L           | T          |
| ARF-ADWIN                               | 98.08        | 98.38        | 99.06      | 98.99        | 0.23        | 4348       |
| ARF-DDM                                 | 98.33        | 98.63        | 99.03      | 99.51        | 0.28        | 3571       |
| SRP-ADWIN                               | 97.44        | 98.00        | 99.30      | 98.65        | 2.85        | 351        |
| SRP-DDM                                 | 99.31        | 99.40        | 99.87      | 99.63        | 2.94        | 440        |
| HT                                      | 95.45        | 95.91        | 99.42      | 97.63        | 0.11        | 9091       |
| KNN-ADWIN                               | 98.61        | 98.83        | 99.70      | 99.27        | 1.24        | 806        |
| PWPAE                                   | 99.04        | 99.05        | 99.94      | 99.63        | 2.80        | 357        |
| <b>MABTSE</b>                           | <b>99.50</b> | <b>99.47</b> | <b>100</b> | <b>99.74</b> | <b>2.75</b> | <b>364</b> |

TABLE 3. Proposed online ddos attack detection model comparison using CICIoT2023 dataset.

| Models comparison using CICIoT2023 dataset |              |              |              |              |             |            |
|--------------------------------------------|--------------|--------------|--------------|--------------|-------------|------------|
|                                            | A            | P            | R            | F1           | L           | T          |
| ARF-ADWIN                                  | 98.60        | 97.58        | 93.32        | 95.36        | 0.25        | 4001       |
| ARF-DDM                                    | 98.74        | 97.30        | 94.39        | 95.83        | 1.20        | 833        |
| SRP-ADWIN                                  | 98.50        | 94.80        | 95.43        | 95.11        | 2.75        | 364        |
| SRP-DDM                                    | 98.56        | 93.97        | 96.83        | 95.38        | 2.41        | 415        |
| HT                                         | 97.60        | 98.22        | 99.25        | 98.72        | 0.19        | 5263       |
| KNN-ADWIN                                  | 98.65        | 94.53        | 96.77        | 95.64        | 0.16        | 6250       |
| PWPAE                                      | 98.99        | 97.94        | 95.43        | 96.67        | 1.45        | 690        |
| <b>MABTSE</b>                              | <b>99.28</b> | <b>99.11</b> | <b>99.10</b> | <b>99.76</b> | <b>1.39</b> | <b>719</b> |

Legend: A=Accuracy, P= Precision, R= Recall, F= F1-score, T= Model learning time, L= throughput

algorithm with the Thompson Sampling method. It selects the dynamically optimal model and the weighting algorithm used for the ensemble. The proposed MABTSE model considers SRP-DDM and KNN-ADWIN as the base models because they achieve better accuracy. The accuracies of the SRP-DDM and KNN-ADWIN were 99.31% and 98.61%, respectively. In terms of precision, recall, and F1 score, the experimental results demonstrate that the proposed model achieved 99.47%, 100%, and 99.74%, respectively. These

results demonstrate that the proposed approach has a relatively accurate and precise DDoS attack detection capability compared to other solutions and is robust to concept drifts. In terms of model training time, the proposed model averaged 2.75 millisecond to train the learning model (Table 1). In comparison, the other online adaptive models HT, ARF-ADWIN, ARF-DDM, KNN-ADWIN, and LB, required less training time than the proposed MABTSE approach. The MABTSE outperformed PWPAE, SRP-ADWIN, and SRP-DDM. This is because MABTSE's ability to dynamically selects the optimal classifier or ensemble of classifiers for each incoming traffic request. Additionally, the AUWPAE approach is employed to combine the base learners (classifiers) for the IoT stream analytics. This strategy facilitates seamless integration and enhances adaptability. Furthermore, it sustains the performance of conventional DDoS attack detection systems in non-adversarial attack scenarios by selecting the most appropriate classifiers for each request type.

The proposed online MABTSE model, achieved 719 samples per second throughput, making it better than SRP-DDM, SRPs-ADWIN, and PWPAE for CICIoT2023 dataset. However, its performance was lower than ARF-ADWIN, HTs, and KNN-ADWIN. This finding is also consistent with the results from the IOTID20 dataset.

Figure 4 and Table 1 presents the performance comparison of the proposed MABTSE model against other online adaptive models including ARF-ADWIN, ARF-DDM, SRP-ADWIN, SRP-DDM, HT, KNN-ADWIN, and PWPAE utilizing the CICIoT2023 dataset. The experimental results indicate that the proposed model achieved an average accuracy of 99.28%, surpassing the performance of other methods. Furthermore, in terms of the precision, recall, and F1-score performance metrics, MABTSE achieved 98.88%, 96.53%, and 97.98%, respectively. These results demonstrate that MABTSE outperforms other online adaptive learning methods in the presence of concept drifts.

We calculated the confidence interval and p-value for the accuracy metrics and proposed MABTSE model and other baseline models including ARF-ADWIN, ARF-DDM, HT, KNN-ADWIN, SRP-DDM, SRP-ADWIN, and PWPAE. The results demonstrate superior performance of the proposed model, supported by statistical significance. The MABTSE model, utilizing IoTID20 dataset achieved a mean accuracy of 99.50%, with confidence interval of [99.44%, 99.56%] and p-value of < 0.0001. These result shows significantly outperform the other baseline models. Similarly, MABTSE model achieved a mean accuracy of 99.28%, a confidence interval of [99.21%, 99.35%], and a p-value of < 0.0001 utilizing CICIoT2023 dataset. These finding further highlight the statistical significance of the proposed model in comparison to the baseline models.

## B. MABTSE UNDER ATTACK

The performance of the proposed MABTE online DDoS detection model decreased significantly under the influence of perturbations generated by the PGD, FGSM, C&C, and

BIM methods utilizing IOTID20 and CICIOT2023 datasets. The accuracy of the MABTSE DDoS detection model declined to a range of 38.92% to 67.12% for IOTID20 dataset and to range of 13.08% to 32.38% for CICIOT2023 datasets, respectively. As shown in Tables 3 and 4, among the four adversarial attacks, C&C was the most effective on both the IOTID20 and CICIOT2023 datasets. Furthermore, C&C attacks caused a significant increase in the model learning time, which is 0.7s and 0.5s for the IOTID20 and CICIOT2023 datasets respectively. This is because the C&C algorithm generates sophisticated and dynamic perturbation patterns, leading to increased complexity. Consequently, the presence of C&C attacks significantly increases the model learning time, because the MABTSE DDoS detection model requires additional processing.

### C. RECOVERED

After the implementation of the Resilient Adversarial Detection and Purification (RADP) method to detect and purify adversarial attacks against an online DDoS attack detection system, the system's performance improved significantly. The accuracy, precision, recall, and F1-scores exhibit significant enhancements, closely resemble to those of the original MABTSE DDoS attack-detection model. The performance of the MABTSE DDoS attack detection model enhanced from a ranged of 99.39% to 99.48% for IOTID20 dataset and from a range of 99.39% to 99.48% for CICIOT2023 datasets, respectively. This demonstrates the effectiveness of the MSAAD framework in identifying and differentiating adversarial samples from original streaming data samples.

**TABLE 4. The effect of adversarial and effect of defense mechanism using IOTID20 dataset.**

|              |      | Models comparison using IOTID20 dataset |       |       |       |     |
|--------------|------|-----------------------------------------|-------|-------|-------|-----|
| Original     |      | A                                       | P     | R     | F1    | T   |
| Under attack |      |                                         |       |       |       |     |
|              | FGSM | 60.58                                   | 80.80 | 4.22  | 5.78  | 0.6 |
|              | BIG  | 40.33                                   | 38.91 | 0.37  | 3.55  | 0.4 |
|              | PGD  | 38.44                                   | 33.94 | 0.16  | 2.56  | 0.4 |
|              | C&W  | 32.38                                   | 26.91 | 0.06  | 1.13  | 0.7 |
| Recovered    |      |                                         |       |       |       |     |
|              | FGSM | 99.48                                   | 99.37 | 99.99 | 99.67 | 0.7 |
|              | BIG  | 99.40                                   | 99.41 | 99.89 | 99.64 | 0.3 |
|              | PGD  | 99.39                                   | 99.40 | 99.99 | 99.71 | 0.3 |
|              | C&W  | 99.43                                   | 99.33 | 99.90 | 99.69 | 0.8 |

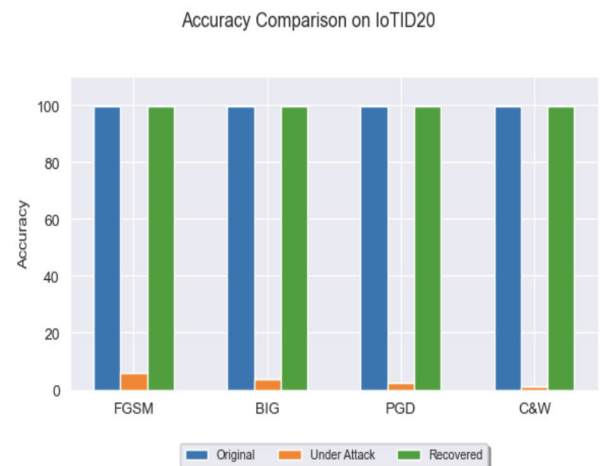
Figure 5 and 6 illustrate the proposed MABTSE model for adversarial attacks, including FGSM, BIG, PGD, and C&W. These experiments demonstrate the vulnerability of the proposed MABTSE online DDoS attack detection models when evaluated using the IOTID20 and CICIOT2023 datasets. Notably, PGD and C&W exhibit particularly strong performance as accuracy declines. The IOTID20 dataset appeared to be more sensitive to FGSM and BIG attacks, showing sharper accuracy declines compared with the CICIOT2023 dataset, which can be attributed to the inherent characteristics of data. However, after implementing the recovery mechanisms (i.e., RADP), the proposed MABTSE model restore performance

**TABLE 5. The effect of adversarial attack and effect of defense mechanism using CICIOT2023 dataset.**

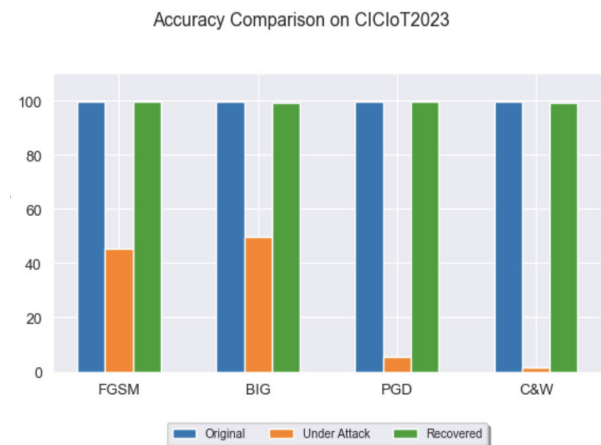
|              |      | Models comparison using CICIOT2023 dataset |       |       |       |     |
|--------------|------|--------------------------------------------|-------|-------|-------|-----|
| Original     |      | A                                          | P     | R     | F1    | T   |
| Under attack |      |                                            |       |       |       |     |
|              | FGSM | 86.20                                      | 96.19 | 31.11 | 45.33 | 0.2 |
|              | BIG  | 84.68                                      | 96.10 | 46.74 | 49.87 | 0.3 |
|              | PGD  | 77.33                                      | 33.32 | 2.22  | 5.48  | 0.3 |
|              | C&W  | 66.60                                      | 23.22 | 1.98  | 1.72  | 0.5 |
| Recovered    |      |                                            |       |       |       |     |
|              | FGSM | 99.26                                      | 99.01 | 99.09 | 99.67 | 0.2 |
|              | BIG  | 99.23                                      | 99.14 | 99.05 | 99.33 | 0.3 |
|              | PGD  | 99.22                                      | 99.11 | 99.11 | 99.71 | 0.3 |
|              | C&W  | 99.27                                      | 99.14 | 99.03 | 99.35 | 0.4 |

Legend: A=Accuracy, P= Precision, R= Recall, F= F1-score, T= Model learning time

to level nearly equivalent to those of original dataset (without adversarial attack). This result demonstrates the robustness of the proposed defence strategy.



**FIGURE 5. Performance change using IOTID20 dataset.**



**FIGURE 6. Performance change using CICIOT2023 dataset.**

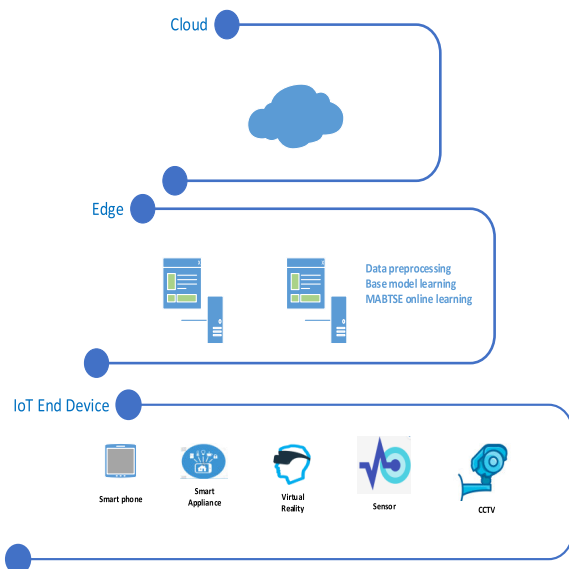
## VI. PROPOSED SOLUTION DEPLOYMENT

This section describes the edge and gateway deployments of the proposed solution. Deploying online IoT DDoS attack

detection at network edges or gateways offers several advantages. The main advantages are the reduced latency and bandwidth consumption. Edges or gateways serve as the points where IoT devices connect to a wide network, as illustrated in Figure 7. Deploying the MABTSE solution at the gateway enables faster detection of DDoS attacks and prevents them from reaching cloud infrastructure or core systems. This local filtering allows for a more rapid response to attacks and helps prevent other network segments from becoming overloaded, thereby reducing latency.

The enhanced security and scalability of edge deployment are two important advantages. Data privacy is maintained and transmission risks are reduced when the data are processed closer to the source. This eliminates the need to send sensitive information to external servers. Gateways are simple to use in various IoT applications owing to their adaptability and ability to support a wide range of IoT devices and protocols. To preserve performance and avoid any single point of failure, an organization can distribute detection capabilities throughout its network by deploying more gateways as needed as the IoT network expands.

Deploying at the edge or gateways to reduce latency and bandwidth usage. This allows quicker response for DDoS attack and preventing the attack reaching to the core infrastructure. The MABTSE model has dynamic capability of classifier selection this ensures efficient resource utilization in the environment where resource is limited. The MABTSE model has better performance using both datasets but it has relatively higher latency for some models. As you can see in the figure 3 and 4, there are three or more concept drifts in a short period of time on the experiment. However, in real-world scenario this would happen less frequently. This shows the proposed MABTSE model is expected to perform better in a real-world. Moreover, the latency of AUWPAE is less than the latency requirement of IoT production applications.



**FIGURE 7.** Deployment architecture for proposed MABTSE solution.

## VII. CONCLUSION AND FUTURE WORK

Adversarial attacks pose a significant challenge to the detection of Distributed Denial of Service (DDoS) attack in IoT systems. These attacks can exploit the vulnerability of online DDoS attack detection solutions, thereby degrade their accuracy. The current adversarial attack detection systems primarily focus on batch learning and are not equipped to handle multiple and unknown adversarial attacks. To address this challenge, this study provides a novel MSAAD framework that can protect online DDoS attack detection solutions against adversarial attacks. The proposed framework comprises of three defense layers: the Resilient Adversarial Detector and Purification (RADP), which detect and purify adversarial attacks against online DDoS attack detection systems for multiple and unknown adversarial attacks; A multiple-classifier approach, which makes it more challenging for an attacker to replicate the DDoS attack detection model; and MABTSE, which select dynamically classifiers or ensembles using the the Multi-Armed Bandits (MAB) algorithm with Thompson Sampling for each incoming traffic request. The frameworks maintain the performance of conventional DDoS attack detection systems in non-adversarial attack scenarios by selecting the most appropriate classifiers for each request type. The experiment results demonstrate the effectiveness of the proposed MABTSE model. The MABTSE achieved an average accuracy of 99.50% and 99.28% on IOTID20 and CICIOT2023 datasets respectively. The accuracy of the DDoS detection model declined significantly under adversarial attacks. however, it is successfully restored to range of 99.39% to 99.48% for IOTID20 dataset and range of 99.01% to 99.14% for CICIOT2023 datasets respectively in adversarial attack scenario. This demonstrates the robustness of MSAAD framework in dynamic IoT environment

Despite the results presented above, this research acknowledges potential limitations of our current study. The scalability of the framework in large-scale IoT networks, characterized by diverse traffic patterns and device heterogeneity, requires further exploration. In future work, we will focus on optimizing the proposed framework and extending its defence capabilities. Additionally, we will address the challenge posed by large scale data and heterogenous IoT network.

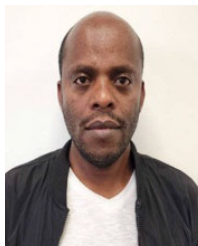
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